ic_autolab round 29 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T09:54:53Z at git `54f00879143f9ed6edb6181369288584449324 05`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-795a16` (`python3 tools/pcc/pcc.py standing PCC-20261104-795a16`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
24	**no_challenger**	none (best candidate `wsum2s-b45-n2`, train S 3.648 / 3.780 / 4.972)	-	-	-	-	-	2.886 / 3.043 / 4.460	1.163 / 1.267 / 2.208	inc: 2.5x / 2.4x / 2.0x
28	**no_challenger**	none (best candidate `wsum2s-b60-n2`, train S 2.588 / 2.846 / 5.161)	-	-	-	-	-	2.296 / 2.398 / 3.317	1.125 / 1.231 / 2.186	inc: 2.0x / 1.9x / 1.5x
32	**no_challenger**	none (best candidate `wsum2s-b90-n2`, train S 2.768 / 3.178 / 6.866)	-	-	-	-	-	2.408 / 2.625 / 4.584	0.932 / 1.023 / 1.846	inc: 2.6x / 2.6x / 2.5x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

- k = 24: no_challenger
- k = 28: no_challenger
- k = 32: no_challenger

Train (development screen, in sample)

k = 24: session `ECBS1h9f5194073f11` (run exit 0, verify exit 0, receipt sha256 `8bfae216e2ebde8f5f92c11912326aa659a1a55e6c5b3c70256caee83290240b`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	- -	`rho.negation`	32	32	1.163 / 1.267 / 2.208	matched reference
`inc`	wsum2g-b30	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=30,free=1`	32	32	2.886 / 3.043 / 4.460	incumbent
`wsum2g-b15-n2`	wsum2g-b15	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=15,free=1`	32	32	4.503 / 4.917 / 8.644	cross-size transfer: wsum2g-b15-n2 is the k = 20 incumbent at N / N_a = 0.062
`wsum2s-b15-n2`	wsum2s-b15	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=15,free=1`	32	32	7.935 / 9.430 / 22.888	exploration: family wsum2s-b15 untried at k = 24, N = 0.0001 x N_a
`wsum2s-b30-n2`	wsum2s-b30	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=30,free=1`	32	32	4.038 / 4.563 / 9.291	exploration: family wsum2s-b30 untried at k = 24, N = 0.0001 x N_a
`wsum2s-b45-n2`	wsum2s-b45	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=45,free=1`	32	32	3.648 / 3.780 / 4.972	exploration: family wsum2s-b45 untried at k = 24, N = 0.0001 x N_a
`wsum2s-b60-n2`	wsum2s-b60	2 `prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=60,free=1`	32	32	5.540 / 5.612 / 6.258	exploration: family wsum2s-b60 untried at k = 24, N = 0.0001 x N_a

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
wsum2s-b90-n2	wsum2s-b90	2	prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=90,free=1	32	32	11.652 / 11.695 / 12.081	exploration: family wsum2s-b90 untried at k = 24, N = 0.0001 x N_a
wsum2s-b120-n2	wsum2s-b120	2	prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=120,free=1	32	32	20.322 / 20.351 / 20.609	exploration: family wsum2s-b120 untried at k = 24, N = 0.0001 x N_a

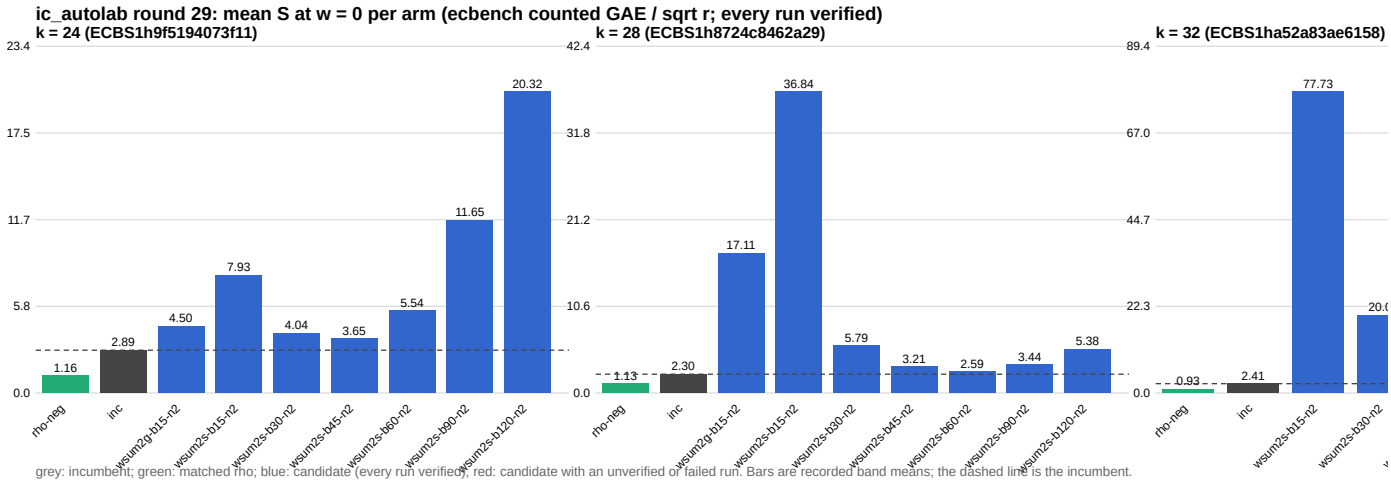
k = 28: session ECBS1h8724c8462a29 (run exit 0, verify exit 0, receipt sha256 4743c51fb2ec9efbbdc27dbc9669d8f6fa96e97afba8eb728dca63c2bb7c8923)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	- -	rho.negation	32	32	1.125 / 1.231 / 2.186	matched reference
inc	wsum2g-b60	2	prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=60,free=1	32	32	2.296 / 2.398 / 3.317	incumbent
wsum2g-b15-n2	wsum2g-b15	2	prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=15,free=1	32	32	17.114 / 18.816 / 34.135	cross-size transfer: wsum2g-b15-n2 is the k = 20 incumbent at N / N_a = 0.062
wsum2s-b15-n2	wsum2s-b15	2	prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=15,free=1	32	32	36.839 / 44.183 / 110.277	exploration: family wsum2s-b15 untried at k = 28, N = 0.0001 x N_a
wsum2s-b30-n2	wsum2s-b30	2	prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=30,free=1	32	32	5.791 / 6.877 / 16.654	exploration: family wsum2s-b30 untried at k = 28, N = 0.0001 x N_a
wsum2s-b45-n2	wsum2s-b45	2	prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=45,free=1	32	32	3.215 / 3.707 / 8.138	exploration: family wsum2s-b45 untried at k = 28, N = 0.0001 x N_a
wsum2s-b60-n2	wsum2s-b60	2	prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=60,free=1	32	32	2.588 / 2.846 / 5.161	exploration: family wsum2s-b60 untried at k = 28, N = 0.0001 x N_a
wsum2s-b90-n2	wsum2s-b90	2	prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=90,free=1	32	32	3.437 / 3.551 / 4.579	exploration: family wsum2s-b90 untried at k = 28, N = 0.0001 x N_a
wsum2s-b120-n2	wsum2s-b120	2	prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=120,free=1	32	32	5.382 / 5.448 / 6.042	exploration: family wsum2s-b120 untried at k = 28, N = 0.0001 x N_a

k = 32: session ECBS1ha52a83ae6158 (run exit 0, verify exit 0, receipt sha256 db2cb6262bc12bd83f7870dc47df8d43b4b3cb4713e55ba428b05210418eb64f)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
rho-neg	rho	- -	rho.negation	32	32	0.932 / 1.023 / 1.846	matched reference
inc	wsum2-b120	2	prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=120,free=1	32	32	2.408 / 2.625 / 4.584	incumbent
wsum2s-b15-n2	wsum2s-b15	2	prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=15,free=1	32	32	77.726 / 93.265 / 233.115	exploration: family wsum2s-b15 untried at k = 32, N = 0.0001 x N_a
wsum2s-b30-n2	wsum2s-b30	2	prime-projected-abscissa:size=2,scan_cap=32	wsum:coef=30,free=1	32	32	20.032 / 24.020 / 59.912	exploration: family wsum2s-b30 untried at k

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`wsum2s-b45-n2`	wsum2s-b45	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=45,free=1`	32	32	10.198 / 12.199 / 30.212	exploration: family wsum2s-b45 untried at k = 32, N = 0.0001 x N_a
`wsum2s-b60-n2`	wsum2s-b60	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=60,free=1`	32	32	5.365 / 6.372 / 15.439	exploration: family wsum2s-b60 untried at k = 32, N = 0.0001 x N_a
`wsum2s-b90-n2`	wsum2s-b90	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=90,free=1`	32	32	2.768 / 3.178 / 6.866	exploration: family wsum2s-b90 untried at k = 32, N = 0.0001 x N_a
`wsum2s-b120-n2`	wsum2s-b120	2	`prime-projected-abscissa:size=2,scan_cap=32`	`wsum:coef=120,free=1`	32	32	2.808 / 3.117 / 5.895	exploration: family wsum2s-b120 untried at k = 32, N = 0.0001 x N_a



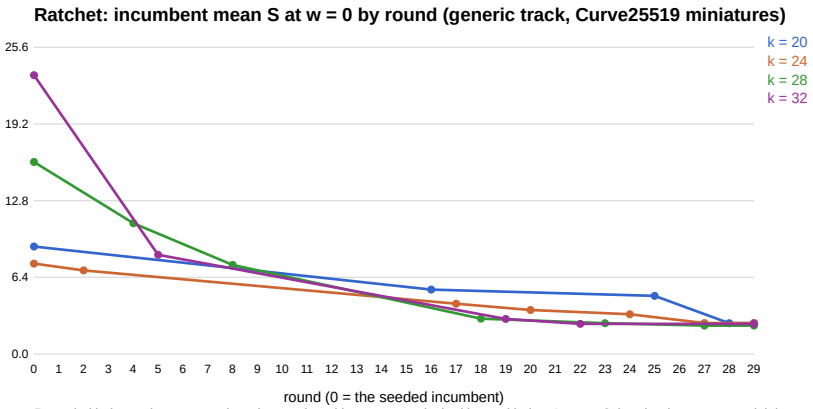
train screen

Selection

- k = 24: no_challenger (Pareto alternative at w = 1: `wsum2s-b45-n2`)
- k = 28: no_challenger (Pareto alternative at w = 1: `wsum2s-b90-n2`)
- k = 32: no_challenger (Pareto alternative at w = 1: `wsum2s-b120-n2`)

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`wsum2g-b15-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=15,free=1`)	28	2.575 / 2.691 / 3.741	1.8x / 1.7x / 1.5x
24	`wsum2g-b30-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=30,free=1`)	27	2.584 / 2.711 / 3.855	2.0x / 2.0x / 1.6x
28	`wsum2g-b60-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=60,free=1`)	27	2.373 / 2.483 / 3.472	2.4x / 2.3x / 1.8x
32	`wsum2-b120-n2` (`prime-projected-abscissa:size=2,scan_cap=32` + `wsum:coef=120,free=1`)	22	2.519 / 2.759 / 4.915	2.7x / 2.7x / 2.7x



ratchet

Evidence and deviations

- ecbench binary sha256: `6229f85b81f9a2e8f2811d675f77f4f830ca536422a1e92680a382183450a62c`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.